

# Kevin Kattakayam

AI/ML Engineer | Generative AI, Multi-Agent Systems, RAG | Internship & Full-Time | Graduating 2027

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## EDUCATION

### Karunya Institute of Technology and Sciences

2023 – 2027 (Expected)

B.Tech, Computer Science and Engineering (AI & ML) – CGPA: 8.39/10

Coimbatore, India

Viswajyothi CMI Public School, Angamaly – Secondary, CBSE (2021): 92.2%; Higher Secondary, CBSE (2023): 86%

## EXPERIENCE

### TheAiSignal – Full Stack AI Developer Intern

May 2026 – Jul 2026

- Engineered AI-powered features across a React frontend, FastAPI backend, and REST API layer in production.
- Maintained deployment pipelines and release workflows across multiple projects using GitHub and Cloudflare.

### Karunya Computer Technology Centre – Generative AI Trainee

Dec 2024 – Present

- Build LLM-powered applications with a team across multiple generative AI projects using LangChain, ChromaDB, and FAISS.
- Apply retrieval-augmented generation, prompt engineering, and OCR-based extraction to document processing workflows.

### YoungMinds Technology Solutions Pvt Ltd – Generative AI Intern

May 2025 – Jul 2025

- Developed generative AI applications and automation tools in Python against LLM and REST APIs.

### Karunya Innovation and Design Studio (KIDS) – Generative AI Intern

May 2025 – Jun 2025

- Built a multi-agent patent drafting and verification system in CrewAI with 5 agents for claim writing, prior-art search, and IPO compliance.
- Generated all 9 IPO-compliant sections with auto-generated diagrams, exporting a filing package (Forms 1, 3, 5) as PDF and DOCX.

### Octanet Services Pvt Ltd – Web Development Intern

May 2024 – Jun 2024

- Built responsive web applications using HTML, CSS, and JavaScript following mobile-first design principles.

## PROJECTS

### PharmaTrace – Pharmaceutical Verification Platform

github.com/KevinKattakayam/pharma\_trace

Python, FastAPI, LangGraph, Groq/Llama 3.3 70B, React 19 PWA, Supabase PostGIS

- Built a 6-agent LangGraph pipeline across 3 drug registries (OpenFDA, RxNav, CDSCO), parallelizing lookups with `asyncio.gather`.
- Enforced strict Pydantic schemas and a field-validator “jargon firewall” against hallucinated terminology, with a SHA-256 audit chain; shipped an offline-first PWA with OCR and four-language voice.

### Document OCR & Structure Extraction Pipeline

github.com/KevinKattakayam/pdf-ocr

Python, OCR, Pydantic, pytest

- Built a production-oriented PDF ingestion pipeline: file validation, OCR for scanned documents, structure extraction with fallbacks, table normalization, and canonical JSON plus Markdown output.
- Enforced Pydantic-validated schemas and added RAG-friendly chunk optimization and CLI batch processing with per-document failure reports.

### V-JEPA Video Representation Learning

github.com/KevinKattakayam/finetunevjpea

PyTorch, V-JEPA, CUDA

- Adapted pretrained V-JEPA self-supervised representations for video classification using attentive probes on UCF101.
- Ran experiments over temporal sampling and probe configurations under constrained GPU memory.

### Schamroth Window Detection – Digital Clubbing Screening

github.com/KevinKattakayam/schamroth

Python, MediaPipe, OpenCV, CLAHE (basis of the IEEE publication)

- Built a computer-vision screening pipeline mapping a gap-presence score to 5 clinical risk levels, evaluated with accuracy, sensitivity, specificity, F1, and a confusion matrix on a 65-image test set.

## TECHNICAL SKILLS

**AI & ML:** LLMs, LLM APIs, Agentic AI, Multi-Agent Systems, RAG, Embeddings, Prompt Engineering, Structured Outputs, NLP

**AI Frameworks:** LangGraph, LangChain, CrewAI, PyTorch, Hugging Face Transformers, Sentence Transformers

**Computer Vision & OCR:** Video Classification, Self-Supervised Learning, MediaPipe, OpenCV, PaddleOCR, Tesseract

**Languages & Backend:** Python, JavaScript, TypeScript, Java, SQL, FastAPI, Streamlit, REST APIs, Pydantic

**Data & Infra:** ChromaDB, FAISS, PostgreSQL, Supabase/PostGIS, Redis, Snowflake, Docker, Git, CI/CD, Cloudflare, Azure

## ACHIEVEMENTS

- IEEE Publication (2026):** “Early Lung Cancer and Heart Disease Prediction from Clubbed Fingers Using Machine Vision”.  
ieeexplore.ieee.org/document/11469562
- Completed Amazon ML Summer School 2026** (Amazon)

## CERTIFICATIONS

Azure AI Engineer Associate, AI-102 (Microsoft) · SQL AI Developer Associate, DP-800 (Microsoft) · Agentforce Specialist (Salesforce) · LLM Security Professional, CLLMSP · SnowPro Associate: Platform (Snowflake) · Oracle Agentic AI Foundations Associate · Oracle OCI AI Foundations Associate